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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

Operational Amplifier

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

100	43	1987 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	38 questions
Marks 2	55 questions
Marks 5	6 questions
Marks 8	1 question
Answer key	at the end

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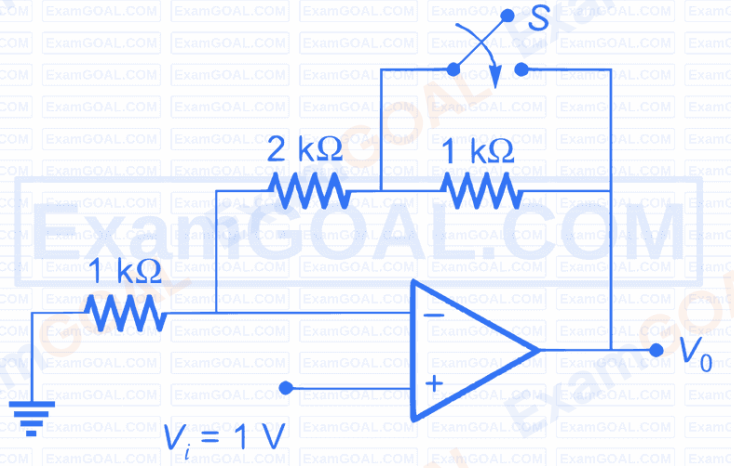
Marks 1 38 questions

Q1. GATE ECE 2026

The ideal OP-AMP circuit shown in the Figure produces output voltage $V_o = x$ when the Switch, S, is open.

Which of the options represents the output voltage when S is closed?

Open solution



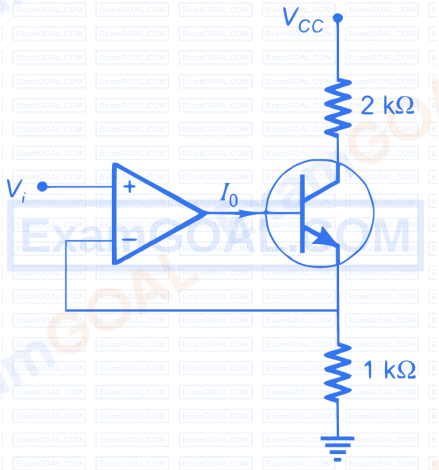
(A) x (B) $\frac{2}{3}x$ (C) $\frac{3}{4}x$ (D) $\frac{1}{2}x$

Q2. GATE ECE 2026

Consider the circuit shown in the Figure with $V_i = 3\ \text{V}$ and $V_{cc} = 12\ \text{V}$. Assume $V_{BE} = 0.7\ \text{V}$ and $\beta_{\text{dc}} = 99$ for the BJT.

Which of the following options is the correct value of the current I_o ?

Open solution



(A) $60\mu\ \text{A}$ (B) $6\mu\ \text{A}$ (C) $3\mu\ \text{A}$ (D) $30\mu\ \text{A}$

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Q3. GATE ECE 2025

Open solution

Which of the following statements is/are TRUE with respect to an ideal opamp?

(A) It has an infinite input resistance

(B) It has an infinite output resistance

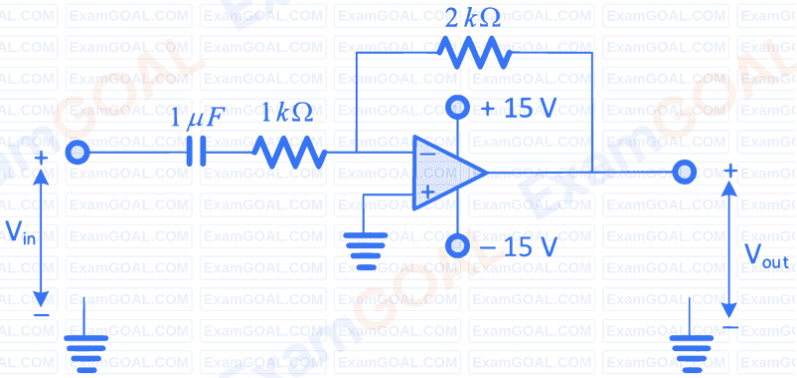
(C) It has an infinite open-loop differential gain

(D) It has an infinite open-loop common-mode gain

Q4. GATE ECE 2022

Open solution

An ideal OPAMP circuit with a sinusoidal input is shown in the figure. The 3 dB frequency is the frequency at which the magnitude of the voltage gain decreases by 3 dB from the maximum value. Which of the options is/are correct?



(A) The circuit is a low pass filter.

(B) The circuit is a high pass filter.

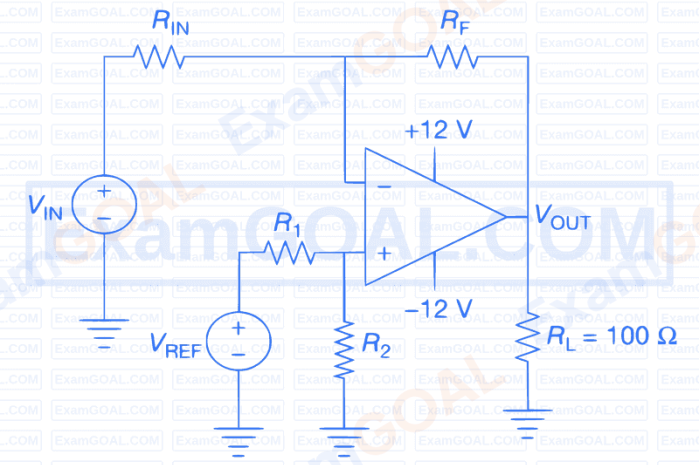
(C) The 3 dB frequency is 1000 rad/s.

(D) The 3 dB frequency is $\frac{1000}{3}$ rad/s.

Q5. GATE ECE 2021

Open solution

For the circuit with an ideal OPAMP shown in the figure. V_{REF} is fixed.



If $V_{OUT} = 1$ volt for $V_{IN} = 0.1$ volt and $V_{OUT} = 6$ volt, for $V_{IN} = 1$ V, where V_{OUT} is measured across R_L connected at the output of this OPAMP. The value of R_F/R_{IN} is

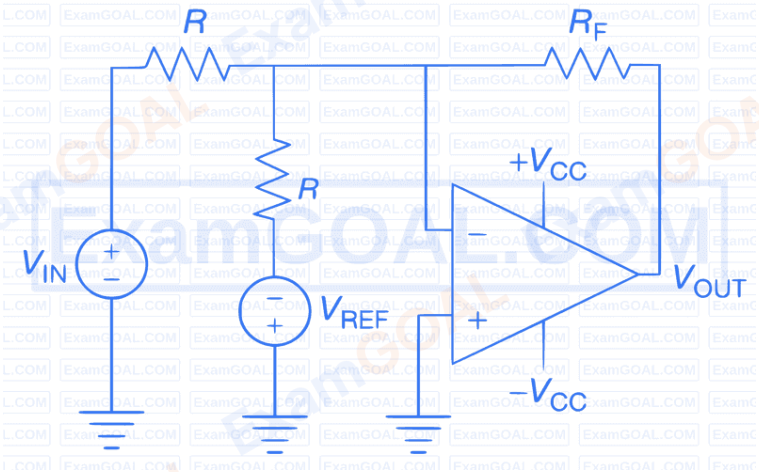
(A) 3.285 (B) 3.825 (C) 2.860 (D) -5.555

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Q6. GATE ECE 2021

Open solution

Consider the circuit with an ideal OPAMP shown in the figure.



Assuming $|V_{IN}| \ll |V_{CC}|$ and $|V_{REF}| \ll |V_{CC}|$. The condition at which V_{OUT} equals to zero is

(A) $V_{IN} = 2V_{REF}$

(B) $V_{IN} = 2 + V_{REF}$

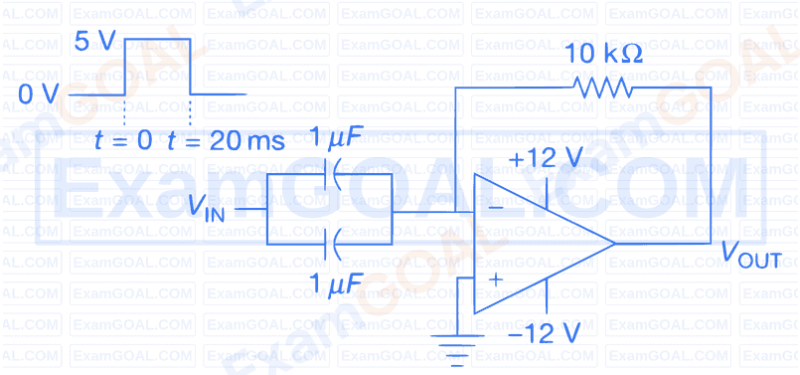
(C) $V_{IN} = V_{REF}$

(D) $V_{IN} = 0.5V_{REF}$

Q7. GATE ECE 2021

Open solution

A circuit with an ideal OPAMP is shown in the figure. A plus V_{IN} of 20 ms duration is applied to the input. The capacitors are initially uncharged.



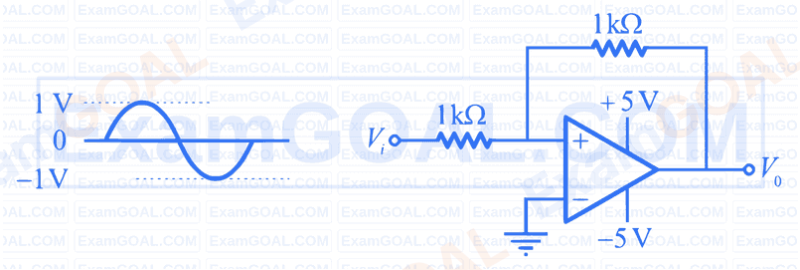
The output voltage V_{OUT} of this circuit at $\tau = 0^+$(in integer) is ____ V.

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Q8. GATE ECE 2020

The components in the circuit shown below are ideal. If the op-amp is in positive feedback and the input voltage V_i is a sine wave of amplitude 1 V , the output voltage V_0 is

Open solution



(A) a constant of either +5 V or -5 V .

(B) a non-inverted sine wave of 2 V amplitude.

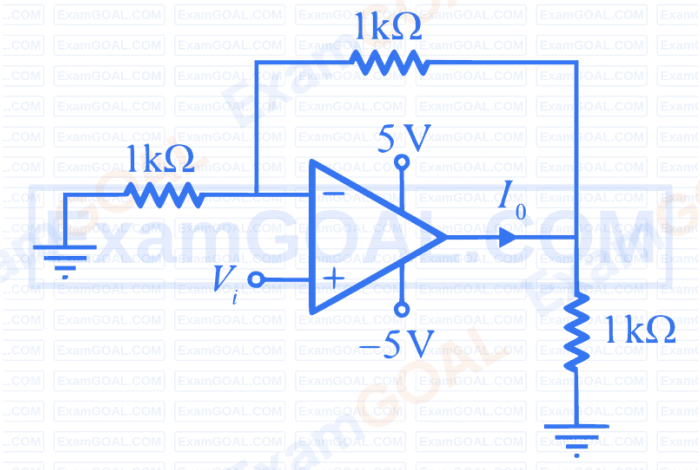
(C) an inverted sine wave of 1 V amplitude.

(D) a square wave of 5 V amplitude.

Q9. GATE ECE 2020

In the circuit shown below, all the components are ideal. If V_i is +2 V , the current I_0 sourced by the OpAmp is ____ mA .

Open solution

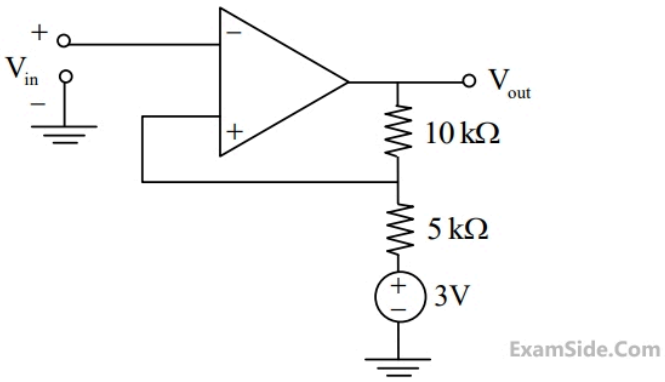


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Q10. GATE ECE 2017 Set 1

For the operational amplifier circuit shown, the output saturation voltages are $\pm 15V$. The upper and lower threshold voltages for the circuit are, respectively.

Open solution

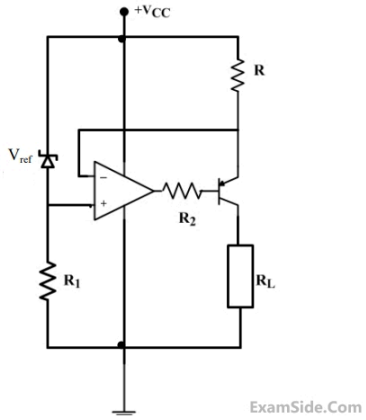


(A) +5 V and -5V (B) +7 V and -3V (C) +3V and -7V (D) +3V and -3V

Q11. GATE ECE 2016 Set 1

Consider the constant current source shown in the figure below. Let β represent the current gain of the transistor

Open solution



The load current I_0 through R_L is

(A) $I_0 = \left[\frac{\beta+1}{\beta}\right]\frac{V_{ref}}{R}$ (B) $I_0 = \left[\frac{\beta}{\beta+1}\right]\frac{V_{ref}}{R}$

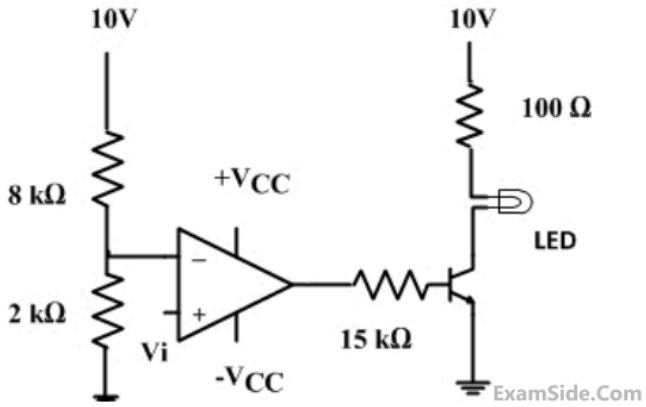
(C) $I_0 = \left[\frac{\beta+1}{\beta}\right]\frac{V_{ref}}{2R}$ (D) $I_0 = \left[\frac{\beta}{\beta+1}\right]\frac{V_{ref}}{2R}$

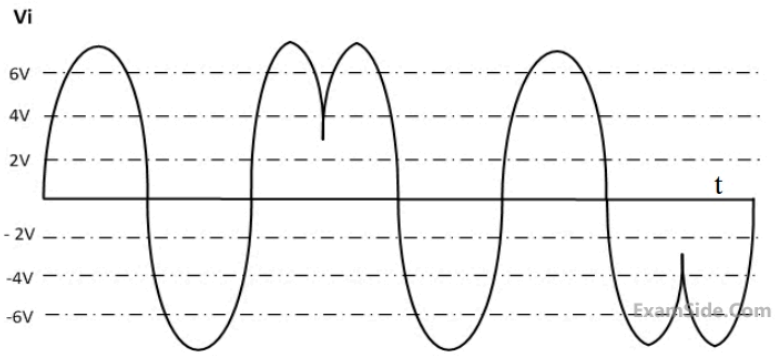
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Q12. GATE ECE 2016 Set 1

Open solution

The following signal V_i of peak voltage 8V is applied to the non-inverting terminal of an ideal Opamp. The transistor has V_{BE} = 0.7 V, $\beta = 100$; V_{LED} = 1.5 V, V_{CC} = 10V and $-V_{CC}$ = -10V.



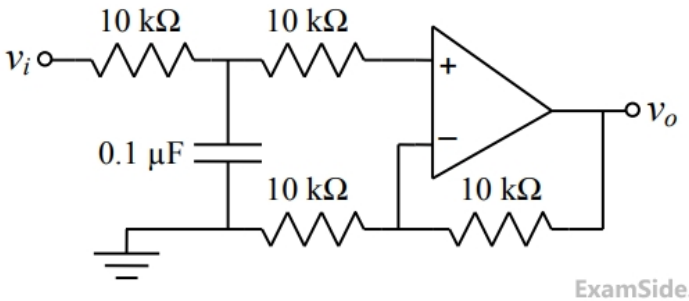


The number of times the LED glows is _____.

Q13. GATE ECE 2015 Set 3

Open solution

In the circuit shown using an ideal op-amp, the 3-dB cut-off frequency (in Hz) is _____.



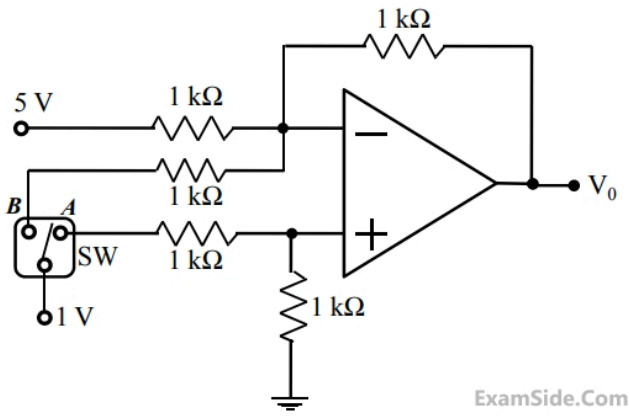
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Q14. GATE ECE 2015 Set 2

In the circuit shown, $V_O = V_{OA}$ for switch SW in position A and $V_0 = V_{OB}$ for SW in position B. Assume that the opamp is ideal.

The value of $\frac{V_{OB}}{V_{OA}}$ is _____

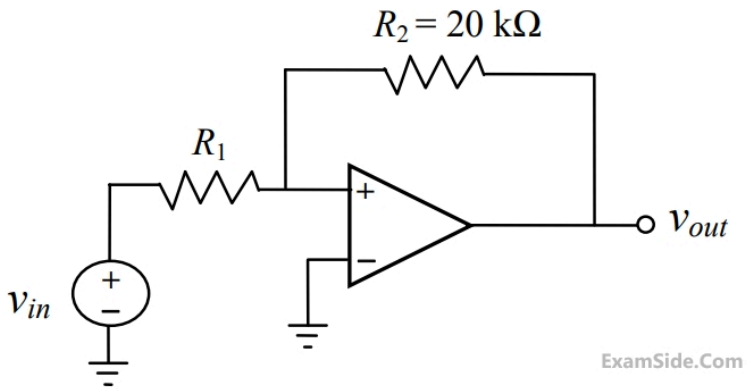
Open solution



Q15. GATE ECE 2015 Set 2

In the bistable circuit shown, the ideal opamp has saturation levels of $\pm 5V$. The value of R_1 (in kΩ) that gives a hystersis width of 500 mV is _____

Open solution

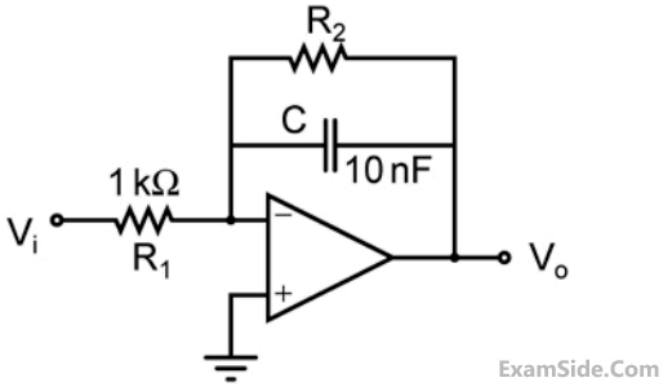


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Q16. GATE ECE 2014 Set 1

In the low - pass filter shown in the figure, for a cut - off frequency of 5 KHz, the value of R_2 (in kΩ) is _____

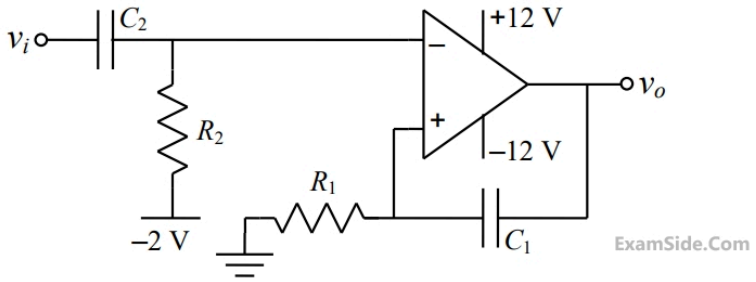
Open solution



Q17. GATE ECE 2014 Set 4

The circuit shown represents

Open solution



(A) a bandpass filter

(B) a voltage controlled oscillator

(C) an amplitude modulator

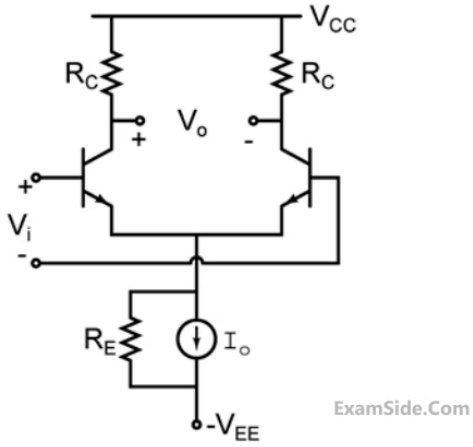
(D) a monostable multivibrator

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Q18. GATE ECE 2014 Set 2

Open solution

In the differential amplifier shown in the figure, the magnitudes of the common-mode and differential-mode gains are A_{cm} and A_d, respectively. If the resistance R_E is increased, then



(A) A_{cm} increases.

(B) common-mode rejection ratio increases.

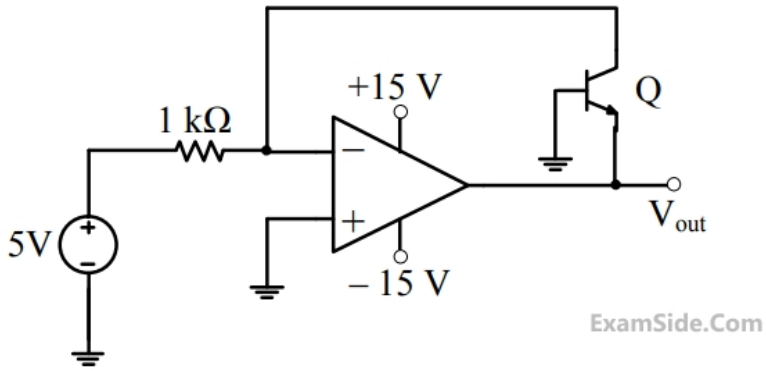
(C) A_d increases.

(D) common-mode rejection ratio decreases.

Q19. GATE ECE 2013

Open solution

In the circuit shown below, what is the output voltage (v_{out}) if a silicon transistor Q and an ideal op-amp are used?



(A) -15 V

(B) -0.7 V

(C) +0.7 V

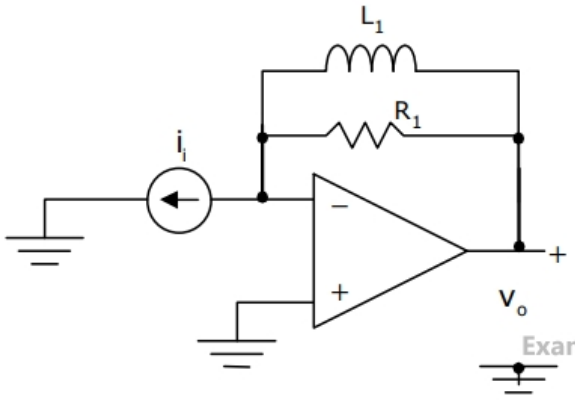
(D) +15 V

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Q20. GATE ECE 2011

The circuit below implements a filter between the input current i_i and the output voltage v_0. Assume that the op-amp- is ideal, The filter implemented is a

Open solution



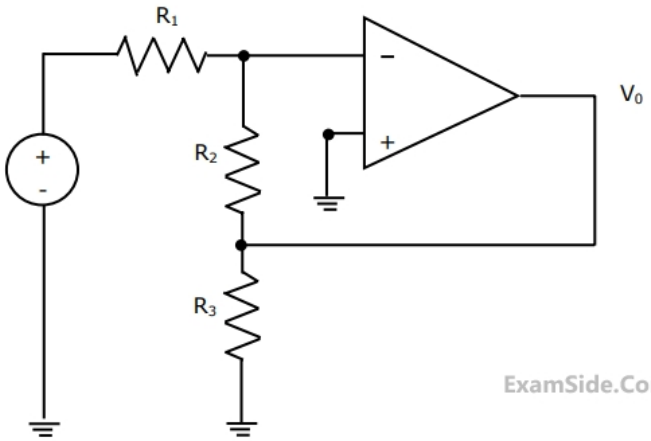
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(A) low pass filter (B) band pass filter (C) band stop filter (D) high pass filter

Q21. GATE ECE 2010

Assuming the OP-AMP to be ideal, the voltage gain of the amplifier shown below is

Open solution



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(A) $-\frac{R_2}{R_1}$ (B) $-\frac{R_3}{R_1}$

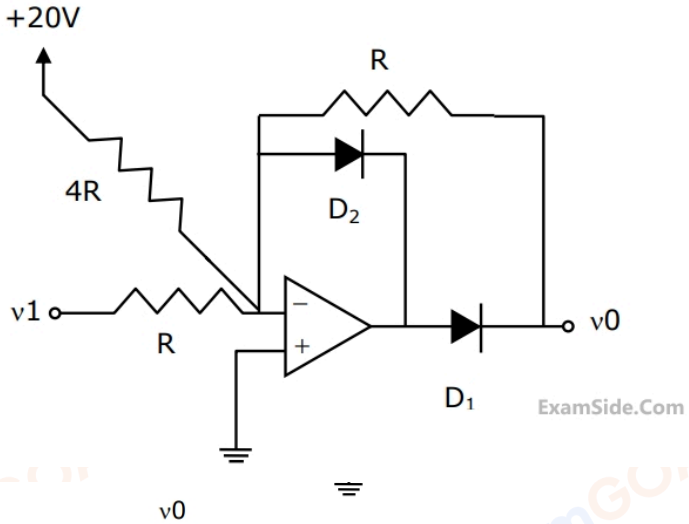
(C) $-\left[\frac{R_2 \| R_3}{R_1}\right]$ (D) $-\left[\frac{R_2+R_3}{R_1}\right]$

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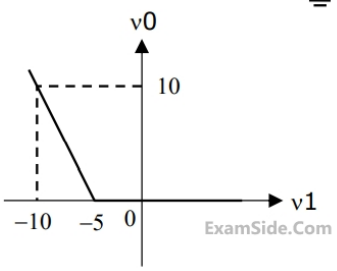
Q22. GATE ECE 2010

Open solution

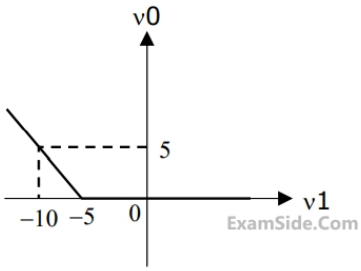
The transfer characteristic for the precision rectifier circuit shown below is (assume ideal OP-AMP and practical diodes)



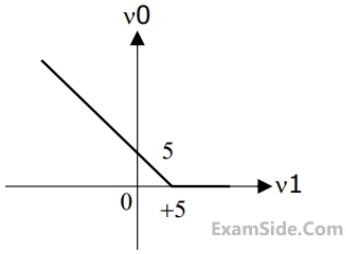
(A)



(B)



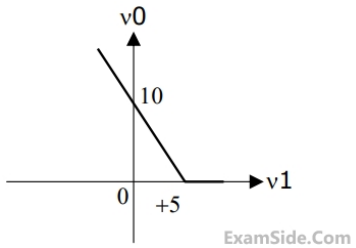
(C)



V

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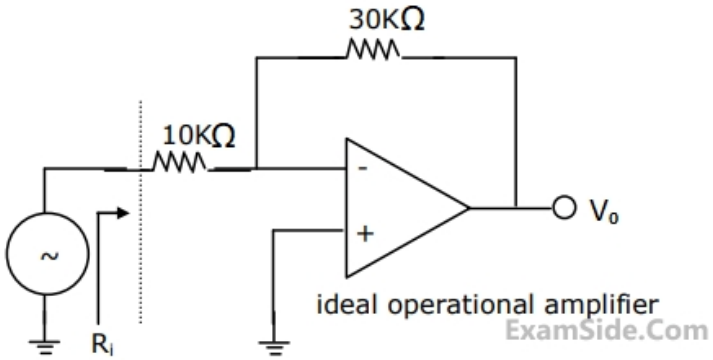
(D)



Q23. GATE ECE 2005

The input resistance R_i of the amplifier shown in the figure is

Open solution



(A) 30/4 KΩ (B) 10 KΩ (C) 40 KΩ (D) infinite

Q24. GATE ECE 2004

An ideal Op-Amp is an ideal

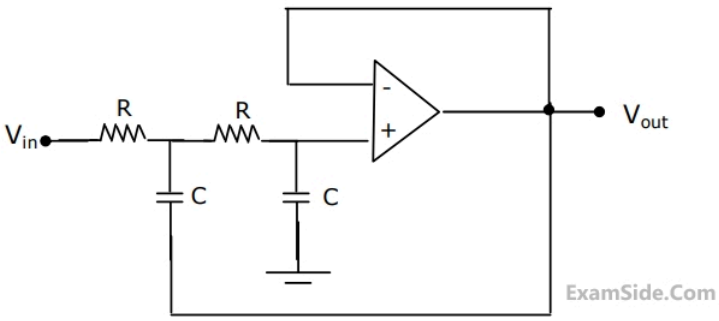
(A) voltage controlled current source
(B) voltage controlled voltage source
(C) current controlled current source
(D) current controlled voltage source

Open solution

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Q25. GATE ECE 2004

The circuit in the figure is a



Open solution

(A) low-pass filter
(B) high-pass filter
(C) band-pass filter
(D) band-reject filter

Q26. GATE ECE 2003

If the differential voltage gain and the common mode voltage gain of a differential Amplifier are 48 dB and 2 dB respectively, then it's common mode Rejection Ratio is

(A) 23 dB
(B) 25 dB
(C) 46 dB
(D) 50 dB

Open solution

Q27. GATE ECE 2001

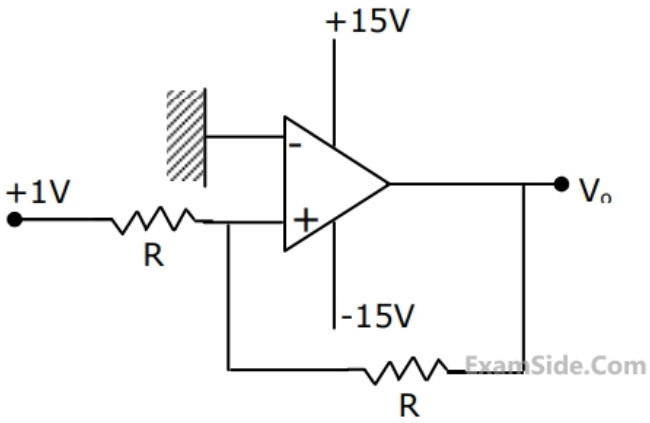
The ideal Op-Amp has the following characteristics.

(A) $R_i = \infty,\ A = \infty,\ R_0 = 0$
(B) $R_i = 0,\ A = \infty,\ R_0 = 0$
(C) $R_i = \infty,\ A = \infty,\ R_0 = \infty$
(D) $R_i = 0,\ A = \infty,\ R_0 = \infty$

Open solution

Q28. GATE ECE 2000

In the circuit of figure V_0 is



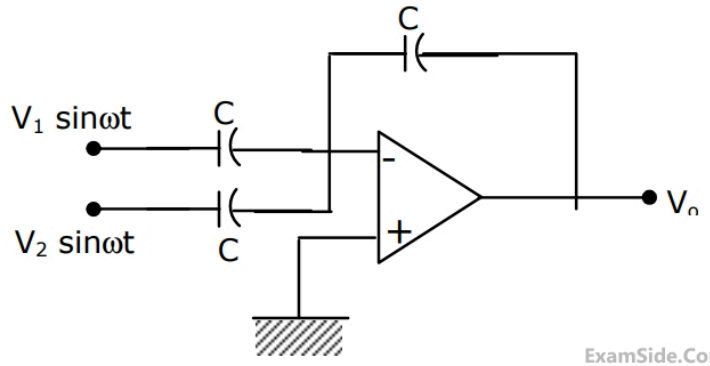
Open solution

(A) -1V
(B) 2 v
(C) +1 V
(D) +15 V

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Q29. GATE ECE 2000

If the Op-Amp in the figure, is ideal, than V_0 is



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Open solution

(A) zero

(B) $(V_1 - V_2) \sin \omega t$

(C) $-(V_1 + V_2) \sin \omega t$

(D) $(V_1 + V_2) \sin \omega t$

Q30. GATE ECE 1999

The first dominant pole encountered in the frequency response of a compensated op-amp is approximately at

(A) 5 Hz (B) 10 KHz (C) 1 MHz (D) 100 MHz

Open solution

Q31. GATE ECE 1998

One input terminal of high gain comparator circuit is connected to ground and a sinusoidal voltage is applied to the other input. The output of comparator will be

(A) a sinusoid

(B) a full rectified sinusoid

(C) a half rectified sinusoid

(D) a square wave

Open solution

Q32. GATE ECE 1998

In a differential Amplifier, CMRR can be improved by using an increased

(A) Emitter resistance

(B) Collector resistance

(C) Power supply voltage

(D) Source resistance

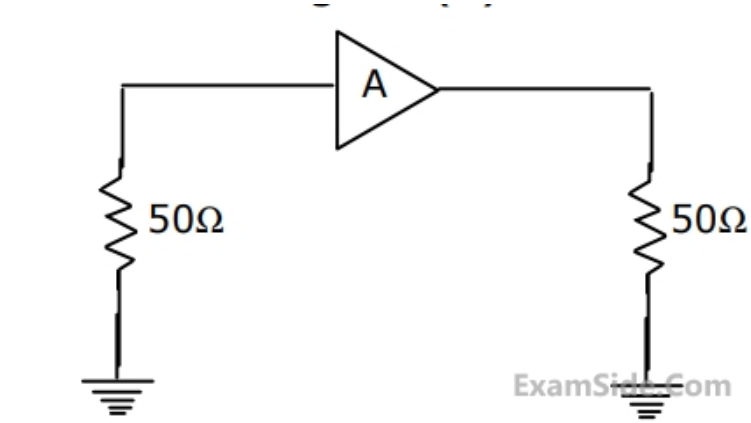
Open solution

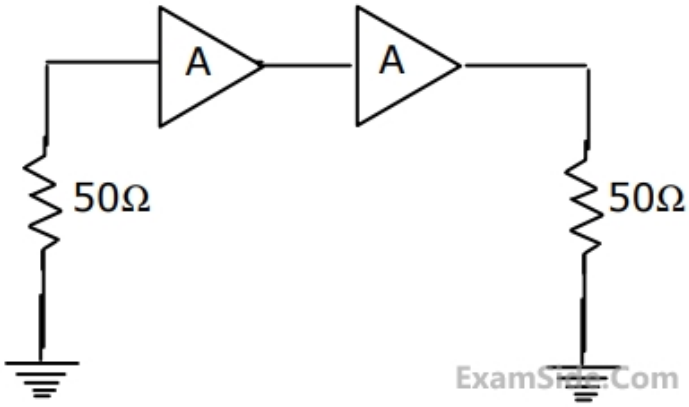
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Q33. GATE ECE 1997

An Amplifier a has 6 dB gain and 50 Ω input and output impedances. The noise figure of this Amplifier as shown in the figure is 3 dB. A cascade of two such Amplifiers as in the figure will have a noise figure of

Open solution



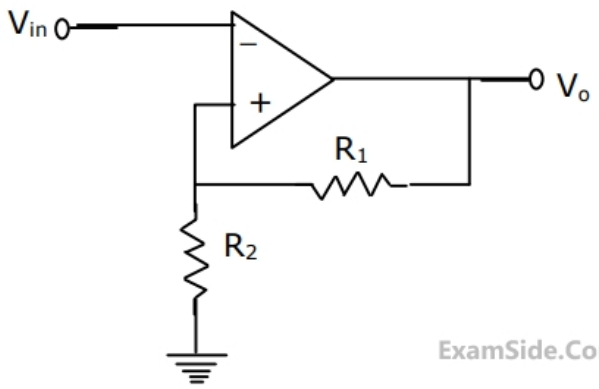


(A) 6 dB
(B) 8 dB
(C) 12 dB
(D) none of the above

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Q34. GATE ECE 1996

The circuit shown in the figure is that of

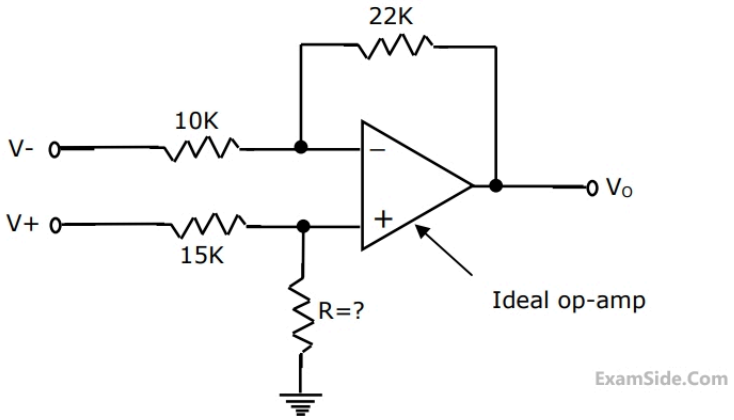


Open solution

(A) a non-inverting Amplifiers
(B) an inverting Amplifier
(C) an oscillator
(D) a Schmitt trigger

Q35. GATE ECE 1995

In the given circuit if the voltage inputs V- and V+ are to be Amplified by the same Amplification factor, the value of 'R' should be



Open solution

Q36. GATE ECE 1995

An op-Amp is used as a zero-crossing detector. If maximum output available from the Op-Amp is $\pm 12\ V_{p-p}$ and the slew rate of the Op-Amp is 12 V/μ sec then the maximum frequency of the input signal that can be applied without causing a reduction in the P-P output is _____

Open solution

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Q37. GATE ECE 1995

A change in the value of the Emitter resistance (R_e in a difference Amplifier.

(A) Affects the difference mode gain A_d

(B) Affects the common mode gain A_c.

(C) Affects both A_d and A_c

(D) Does not affect either A_d (or) A_c.

Open solution

Q38. GATE ECE 1994

The frequency compensation is used in OP-Amps to increase its_____

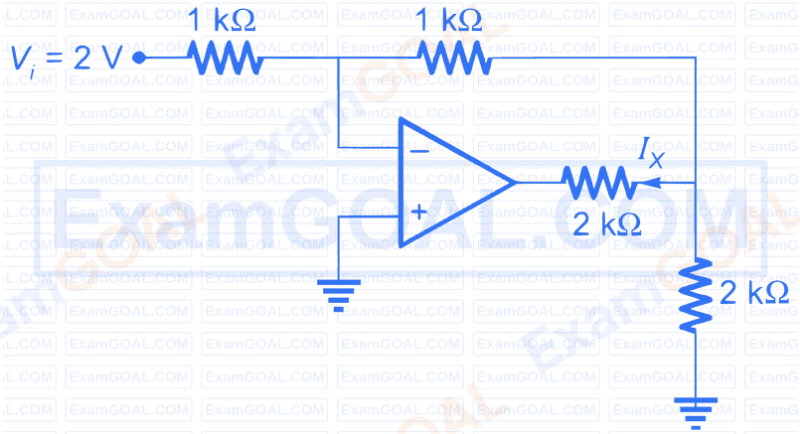
Open solution

Marks 2 55 questions

Q39. GATE ECE 2026

A circuit using an ideal OP-AMP is shown in the Figure.

Which of the following options gives the correct value of the current I_X ?



(A) 2.0 mA

(B) 1.5 mA

(C) 3.0 mA

(D) 0 mA

Open solution

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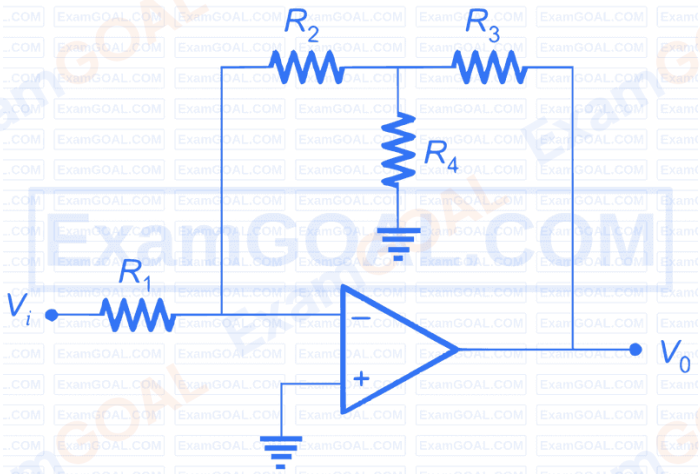
Q40. GATE ECE 2026

Open solution

Consider an ideal OP-AMP circuit as shown in the Figure.

The resistances $R_1 = R_2 = R_3 = R_4 = 50\text{k}\Omega$.

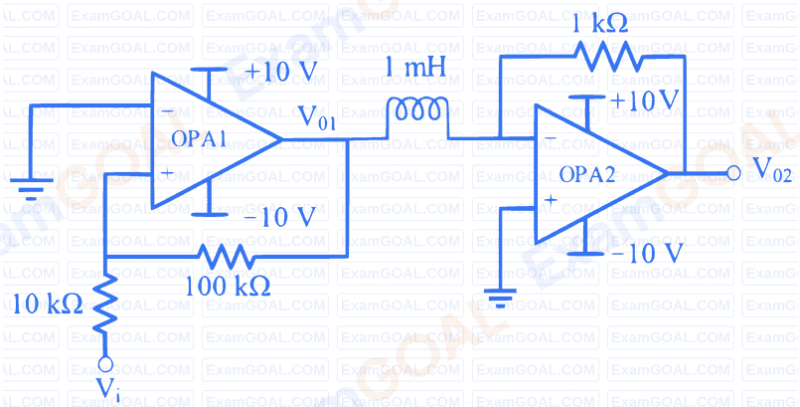
The magnitude of the closed loop gain is ____ . (rounded off to two decimal places)



Q41. GATE ECE 2024

Open solution

The opamps in the circuit shown are ideal, but have saturation voltages of ±10 V.



Assume that the initial inductor current is 0 A. The input voltage (Vi) is a triangular signal with peak voltages of ±2 V and time period of 8 µs. Which one of the following statements is true?

(A) V_{01} is delayed by 2 µs relative to V_i, and V_{02} is a triangular waveform.

(B) V_{01} is not delayed relative to V_i, and V_{02} is a trapezoidal waveform.

(C) V_{01} is not delayed relative to V*i*, and V_{02} is a triangular waveform.

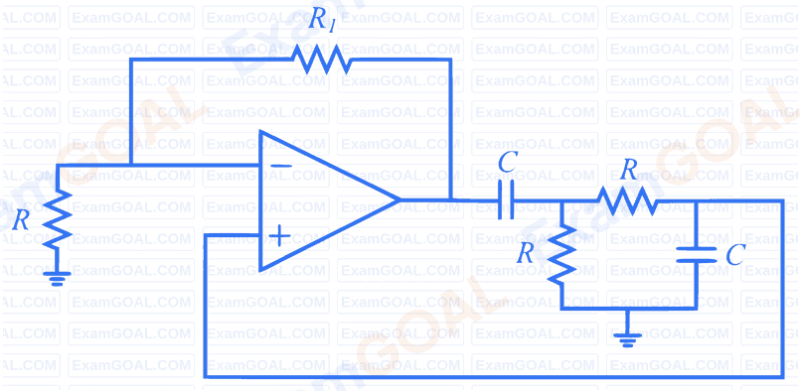
(D) V_{01} is delayed by 1 µs relative to V_i, and V_{02} is a trapezoidal waveform.

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Q42. GATE ECE 2024

Open solution

In the circuit below, the opamp is ideal.



If the circuit is to show sustained oscillations, the respective values of R_1 and the corresponding frequency of oscillation are ____.

(A) $29R$ and $1/(2\pi\sqrt{6}RC)$

(B) $2R$ and $1/(2\pi RC)$

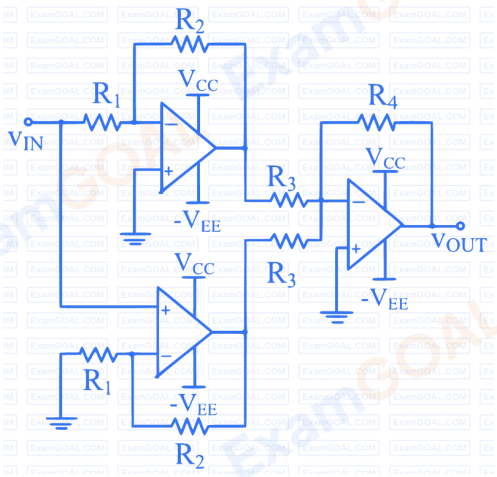
(C) $29R$ and $1/(2\pi RC)$

(D) $2R$ and $1/(2\pi\sqrt{6}RC)$

Q43. GATE ECE 2023

Open solution

The $\frac{V_{OUT}}{V_{IN}}$ of the circuit shown below is



(A) $-\frac{R_4}{R_3}$

(B) $\frac{R_4}{R_3}$

(C) $1+\frac{R_4}{R_3}$

(D) $1-\frac{R_4}{R_3}$

Q44. GATE ECE 2023

Open solution

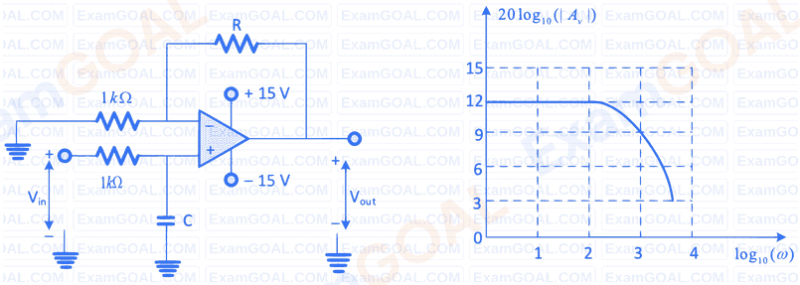
A sample and hold circuit is implemented using a resistive switch and a capacitor with a time constant of 1 μs. The time for the sampling switch to stay closed to charge a capacitor adequately to a full scale voltage of 1 V with 12-bit accuracy is _____ μs (rounded off to two decimal places).

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Q45. GATE ECE 2022

Open solution

A circuit with an ideal OPAMP is shown. The Bode plot for the magnitude (in dB) of the gain transfer function ($A_v(j\omega) = V_{out}(j\omega)/V_{in}(j\omega)$) of the circuit is also provided (here, ω is the angular frequency in rad/s). The values of R and C are _____.



(A) R = 3 kΩ, C = 1 μF

(B) R = 1 kΩ, C = 3 μF

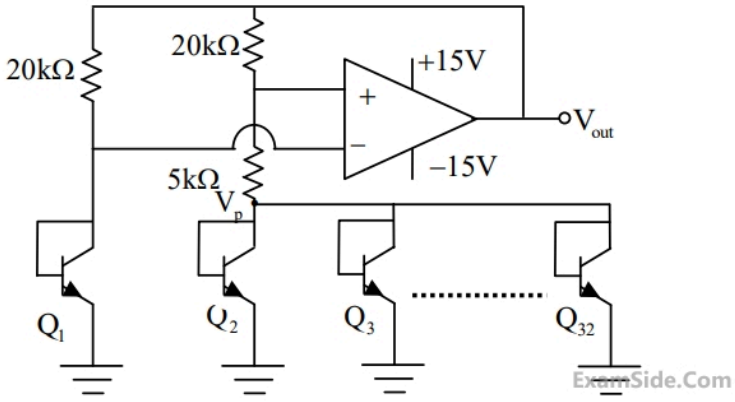
(C) R = 4 kΩ, C = 1 μF

(D) R = 3 kΩ, C = 2 μF

Q46. GATE ECE 2017 Set 2

Open solution

In the voltage reference circuit shown in the figure, the op-amp is ideal and the transistors Q_1, Q_2,....., Q_{32} are identical in all respects and have infinitely large values of common-emitter current gain β. The collector current (I_C) of the transistors is related to their base emitter voltage (V_{BE}) by the relation $I_C = I_S \exp(V_{BE}/V_T)$; where I_s is the saturation current. Assume that the voltage V_P shown in the figure is 0.7 V and the thermal voltage V_T=26mV



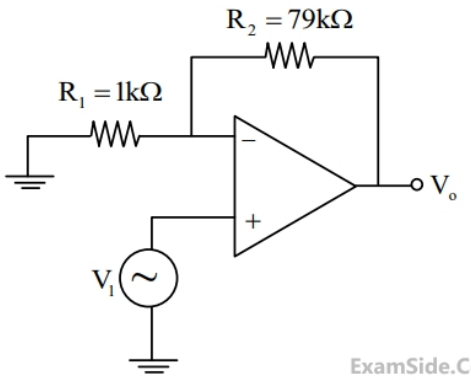
The output voltage V_{out} (in volts) is _____.

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Q47. GATE ECE 2017 Set 1

Open solution

The amplifier circuit shown in the figure is implemented using a compensated operational amplifier (op-amp), and has an open-loop voltage gain, A_0 10^5 V/V and an open-loop cut-off frequency, f_C = 8 Hz. The voltage gain of the amplifier at 15 kHz, in V/V, is _____.

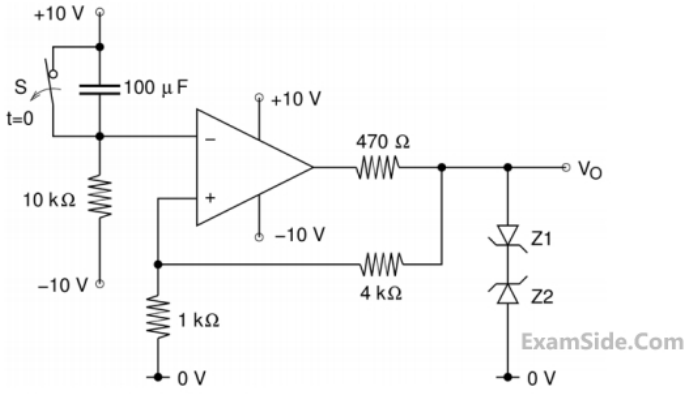


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Q48. GATE ECE 2016 Set 2

Open solution

In the opamp circuit shown, the Zener diodes Z1 and Z2 clamp the output voltage V_o to +5 V or -5 V. The switch S is intially closed and is opened at time t = 0



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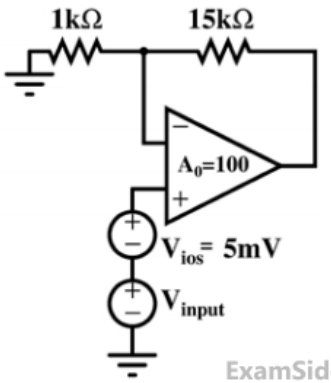
The time t = t_1 (in seconds) at which V_o changes state is _____.

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Q49. GATE ECE 2016 Set 2

Open solution

An opamp has a finite open loop voltage gain of 100. Its input offset voltage V_{ios} (= +5mV) is modeled as shown in the circuit below. The amplifier is ideal in all other respects. V_{input} is 25 mV.



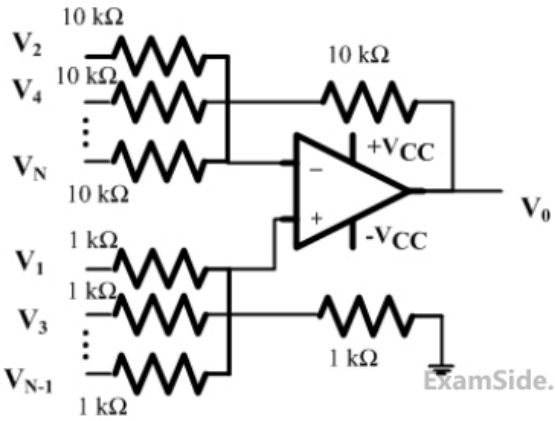
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The output voltage (in millivolts) is _____.

Q50. GATE ECE 2016 Set 1

Open solution

An ideal opamp has voltage source V_1, V_3, V_5, ..., V_{N-1}, connected to the non- inverting input and V_2, V_4, V_6, ...V_N connected to the inverting input as shown in the figure below ($+V_{CC}$= 15 volt, $-V_{CC}$ = -15 volt). The voltage V_1, V_2, V_3,V_4, V_5, V_6 ..... are 1, -1/2, 1/3, -1/4, 1/5, -1/6, ... volt, respectively. As N approaches infinity, the output voltage (in volt) is _____



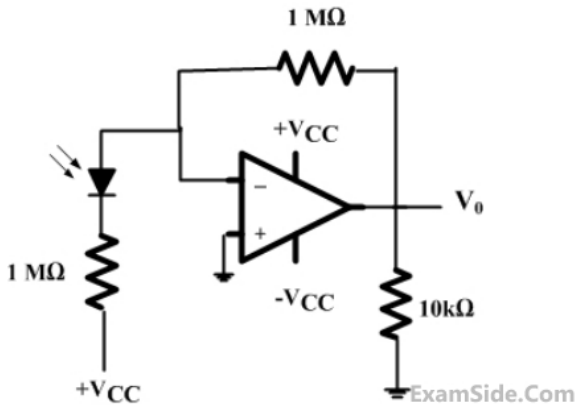
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Q51. GATE ECE 2016 Set 1

Open solution

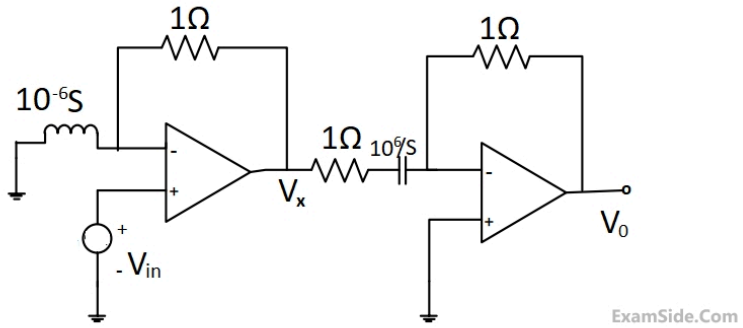
A p-i-n photo diode of responsivity 0.8A/W is connected to the inverting input of an ideal opamp as shown in the figure, $+V_{CC}$ = 15V, - V_{CC} = -15V, Load resistor R_L = 10 kΩ . If 10μW of power is incident on the photodiode, then the value of the photocurrent (in μA) through the load is _____.



Q52. GATE ECE 2016 Set 3

Open solution

For the circuit shown in the figure, $R_1 = R_2 = R_3$ = 1 Ω, L = 1 μH and C = 1 μF. If the input V_{in} = cos(10^6t) , then the overall voltage gain (V_{out} /V_{in}) of the circuit is _____.

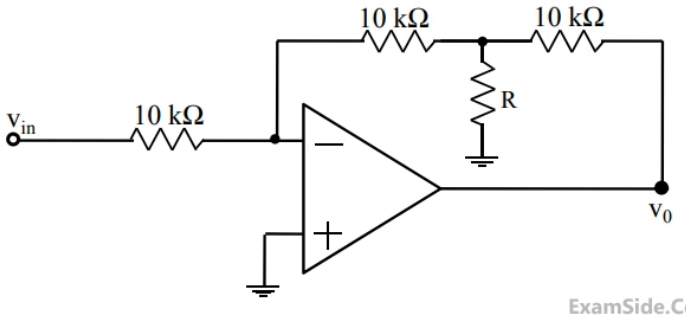


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Q53. GATE ECE 2015 Set 3

In the circuit shown, assume that the opamp is ideal. If the gain (V_0/V_{in}) is -12, the value of R (in KΩ) is _____.

Open solution

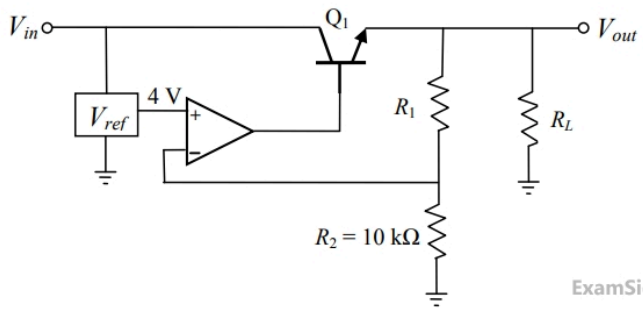


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Q54. GATE ECE 2015 Set 2

For the voltage regulator circuit shown, the input voltage (V_{in}) is 20V ± 20% and the regulated output voltage (V_{out}) is 10 V. Assume the opamp to be ideal . For a load R_L drawing 200 mA, the maximum power dissipation in Q_1 (in Watts) is _____.

Open solution



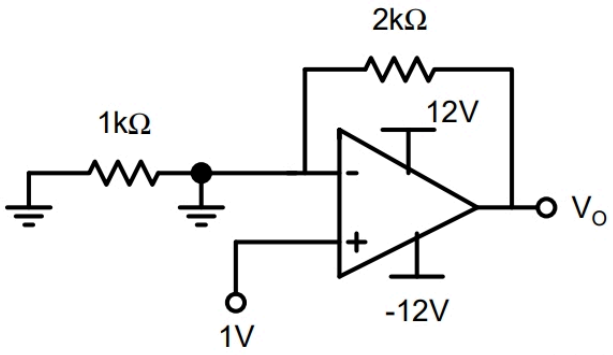
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Q55. GATE ECE 2015 Set 2

Assuming that the op-amp in the circuit shown below is ideal, the output voltage V_0 (in volts) is _____.

Open solution

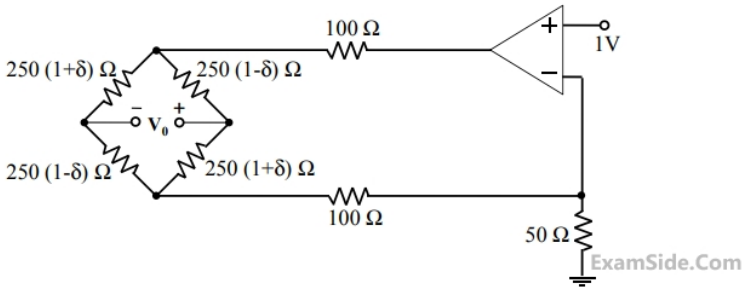


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Q56. GATE ECE 2015 Set 1

In the circuit shown, assume that the opamp is ideal. The bridge output voltage V_0 (in mV) for $\delta = 0.05$ is_____.

Open solution



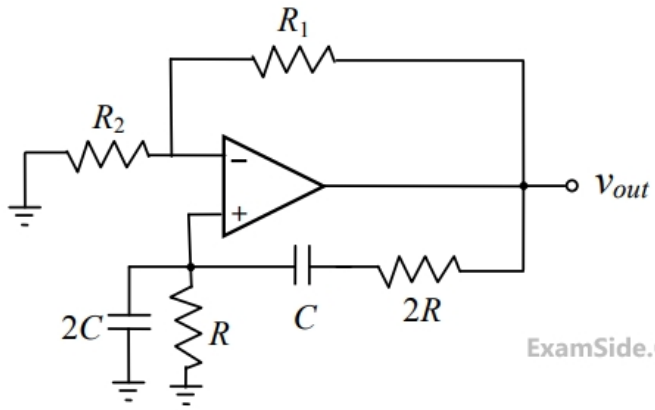
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Q57. GATE ECE 2015 Set 1

The circuit shown in the figure has an ideal opamp. The oscillation frequency and the condition to sustain the oscillations, respectively, are

Open solution



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(A) $\frac{1}{CR}$ and $R_1 = R_2$

(B) $\frac{1}{CR}$ and $R_1 = 4R_2$

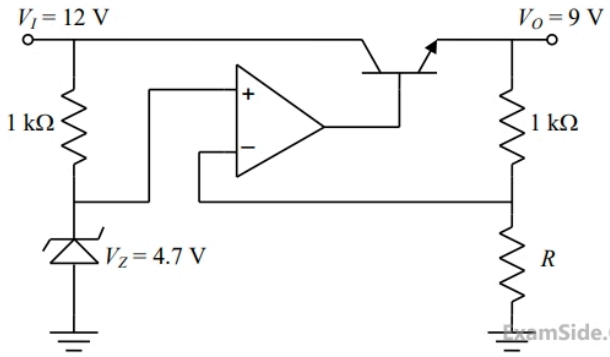
(C) $\frac{1}{2CR}$ and $R_1 = R_2$

(D) $\frac{1}{2CR}$ and $R_1 = 4R_2$

Q58. GATE ECE 2014 Set 1

In the voltage regulator circuit shown in the figure, the op - amp is ideal. The BJT has V_{BE} = 0.7 V and β = 100, and the Zener voltage is 4.7 V. For a regulated output of 9 V, the value of R (in Ω) is _____

Open solution



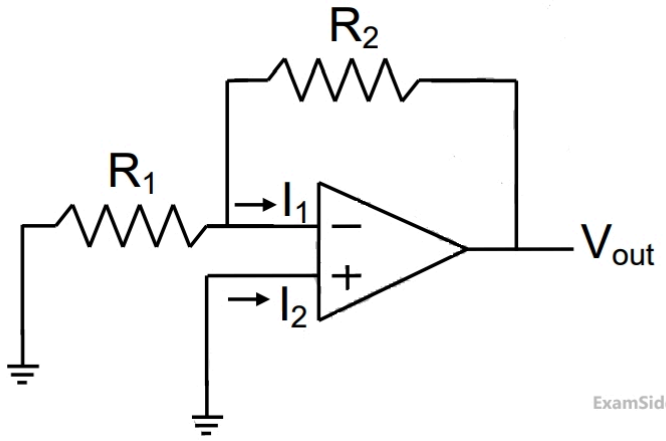
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Q59. GATE ECE 2014 Set 1

In the circuit shown, the op-amp has finite input impedance, infinite voltage gain and zero input offset voltage. The output voltage V_{out} is

Open solution



(A) $-I_2\,(R_1+R_2)$

(B) I_2R_2

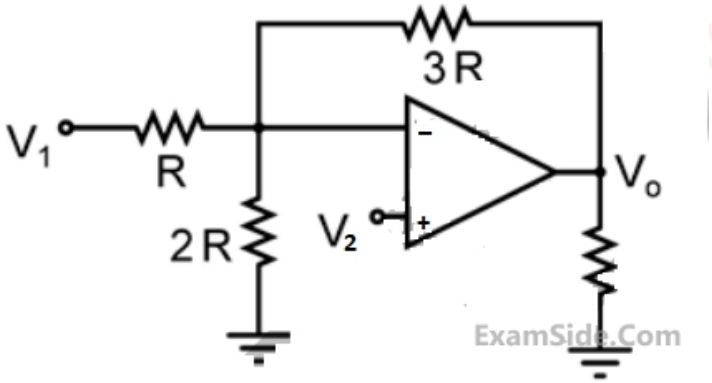
(C) I_1R_2

(D) $-I_1\,(R_1+R_2)$

Q60. GATE ECE 2014 Set 3

Assuming that the Op-amp in the circuit shown is ideal, V_0 is given by

Open solution



(A) $\frac{5}{2}V_1-3V_2$

(B) $2V_1-\frac{5}{2}V_2$

(C) $-\frac{3}{2}V_1+\frac{7}{2}V_2$

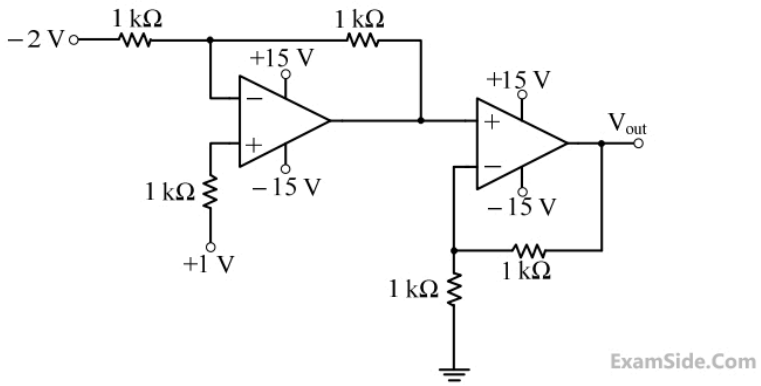
(D) $-3V_1+\frac{11}{2}V_2$

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Q61. GATE ECE 2013

Open solution

In the circuit shown below the op-amps are ideal. Then V_{out} in Volts is

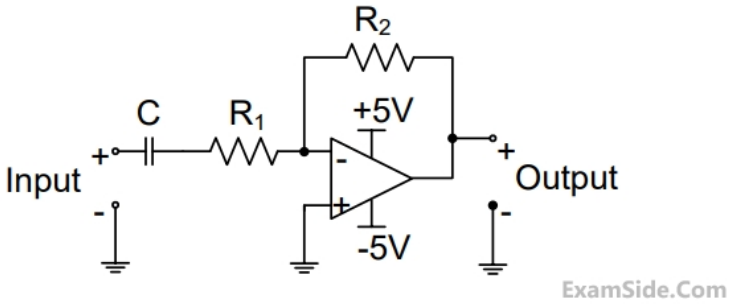


(A) 4 (B) 6 (C) 8 (D) 10

Q62. GATE ECE 2012

Open solution

The circuit shown is a



(A) low pass filter with
$f_3dB = \frac{1}{(R_1+R_2)C} rad/s$

(B) high pass filter with
$f_3dB = \frac{1}{R_1C} rad/s$

(C) low pass filter with
$f_3dB = \frac{1}{R_1C} rad/s$

(D) high pass filter with
$f_3dB = \frac{1}{(R_1+R_2)C} rad/s$

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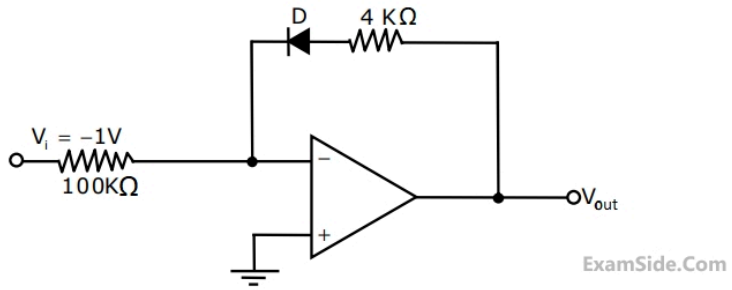
Q63. GATE ECE 2008

Open solution

Consider the following circuit using an ideal Op-Amp. The I-V characteristics of the diode is described by the relation

$$I = I_0\left[e^{\frac{V}{VT}} - 1\right], where\, V_T = 25mV, I_0 = 1\mu A$$

and **V** is the voltage across the diode (taken as positive for forward bias). For an input voltage $V_i = -1V$, the output voltage V_0 is



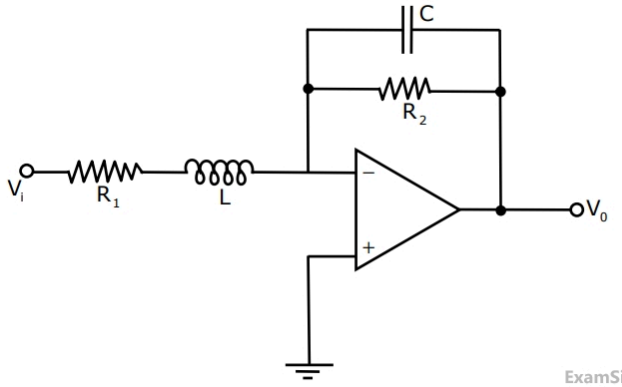
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(A) 0 V (B) 0.1 V (C) 0.7 V (D) 1.1 V

Q64. GATE ECE 2008

Open solution

The Op-Amp circuit shown above represents a



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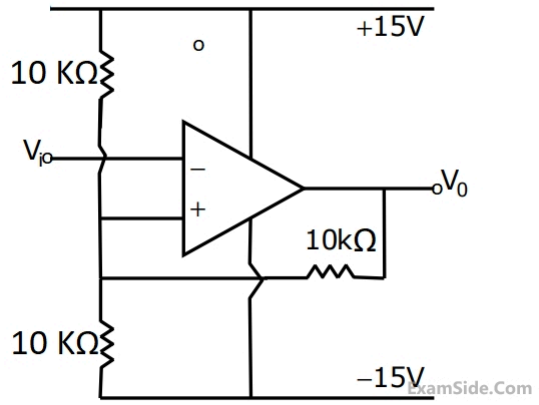
(A) high pass filter (B) low pass filter
(C) band pass filter (D) band reject filter

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Q65. GATE ECE 2008

Open solution

Consider the Schmitt trigger circuit shown below.



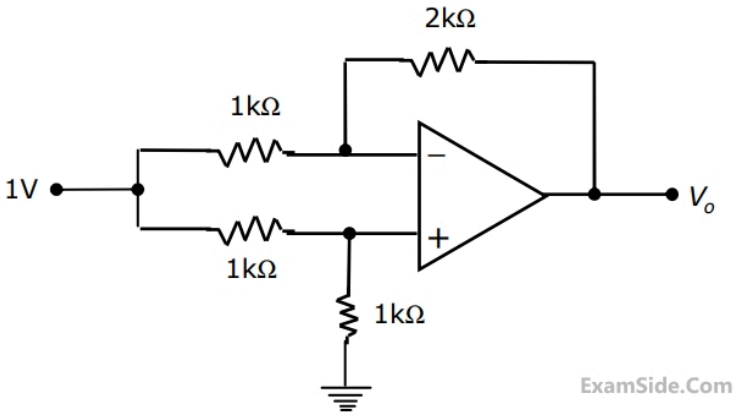
A triangular wave which goes from -12V to 12V is applied to the inverting input of the OP-AMP. Assume that the output of the OP-AMP swings from +15V to -15V. The voltage at the non-inverting input switches between.

(A) -12V and +12V (B) -7.5V and +7.5V (C) -5 V and +5 V (D) 0 V and 5 V

Q66. GATE ECE 2007

Open solution

For the Op-Amp circuit shown in the figure, V_0 is



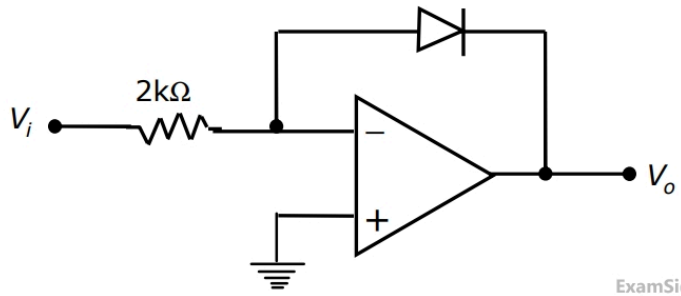
(A) 2 V (B) -1 V (C) -0.5 V (D) 0.5 V

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Q67. GATE ECE 2007

Open solution

In the Op-Amp circuit shown, assume that the diode current follows the equation $I = I_s \exp(V/V_T)$. For $V_i = 2V$, $V_o = V_{01}$, and for $V_i = 4$ V, $V_o = V_{02}$. The relationship between V_{01} and V_{02} is



(A) $V_{02} = \sqrt{2}\,V_{01}$

(B) $V_{02} = e^2 V_{01}$

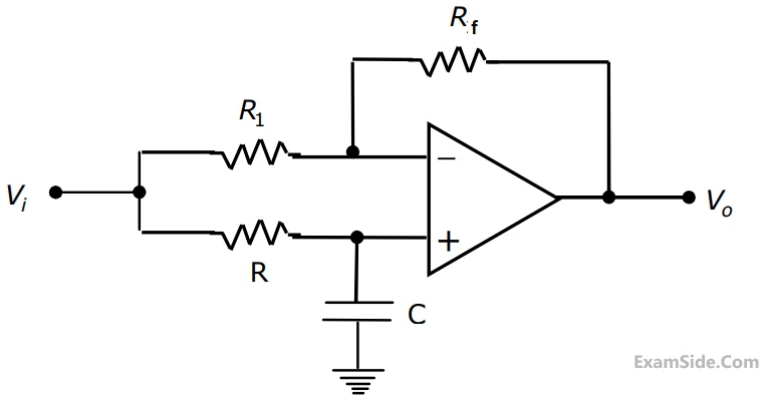
(C) $V_{02} = V_{01} In\, 2$

(D) $V_{01} - V_{02} = V_T In\, 2$

Q68. GATE ECE 2007

Open solution

Consider the OP-Amp circuit shown in the figure.
The transfer function $V_o(s)/V_i(s)$ is



(A) $\frac{1-sRC}{1+sRC}$

(B) $\frac{1+sRC}{1-sRC}$

(C) $\frac{1}{1-sRC}$

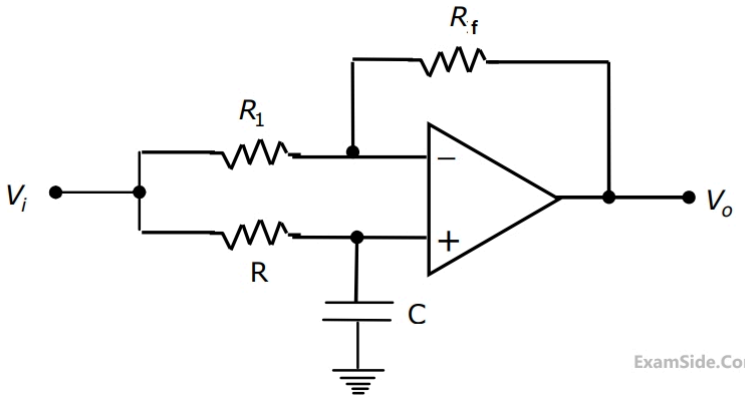
(D) $\frac{1}{1+sRC}$

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Q69. GATE ECE 2007

Open solution

Consider the OP-Amp circuit shown in the figure.



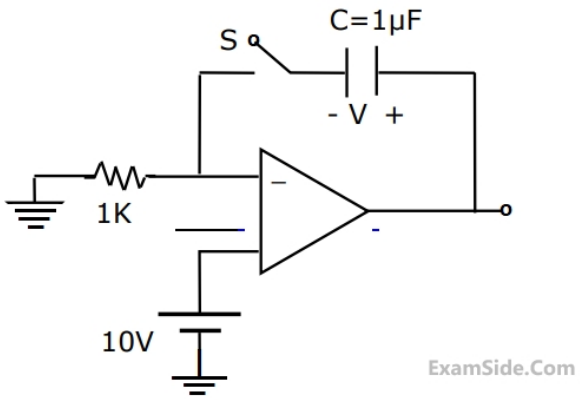
If $V_i = V_1 \sin(\omega\tau + \phi)$, then the minimum and maximum values of ϕ (in radians) are respectively

(A) $-\pi/2$ *and* $\pi/2$ **(B)** 0 and $\pi/2$ **(C)** $-\pi$ and 0 **(D)** $-\pi/2$ and 0

Q70. GATE ECE 2006

Open solution

For the circuit shown in the following figure, the capacitor C is initially uncharged. At t = 0, the switch S is closed. The voltage V_C across the capacitor at t = 1 millisecond is



In the figure shown above, the OP-AMP is supplied with $\pm$ 15V.

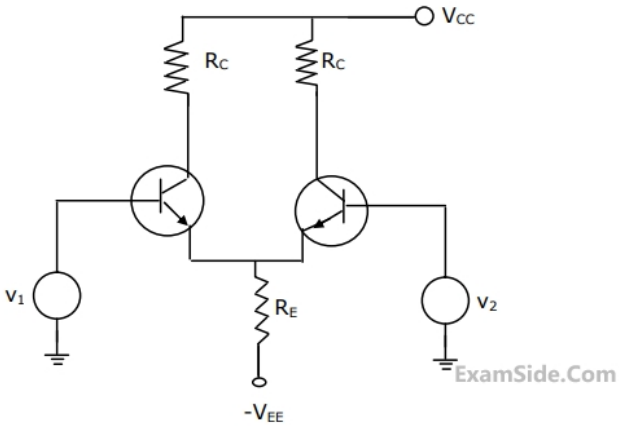
(A) 0 Volts **(B)** 6.3 Volts **(C)** 9.45 Volts **(D)** 10 Volts

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Q71. GATE ECE 2005

In an ideal differential amplifier shown in the figure, a large value of (R_E).

Open solution



(A) increases both the differential and common-mode gains.

(B) increases the common-mode gain only.

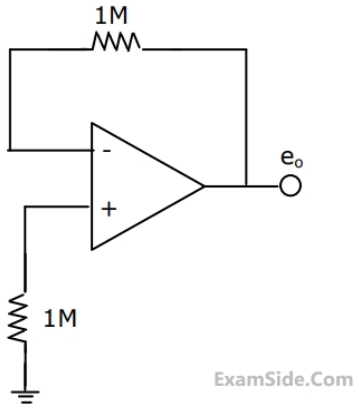
(C) decreases the differential-mode gain only.

(D) decreases the common-mode gain only

Q72. GATE ECE 2005

The voltage e_0 indicated in the figure has been measured by an ideal voltmeter. Which of the following can be calculated?

Open solution



(A) Bias current of the inverting input only.

(B) Bias current of the inverting and non-inverting inputs only.

(C) Input offset current only.

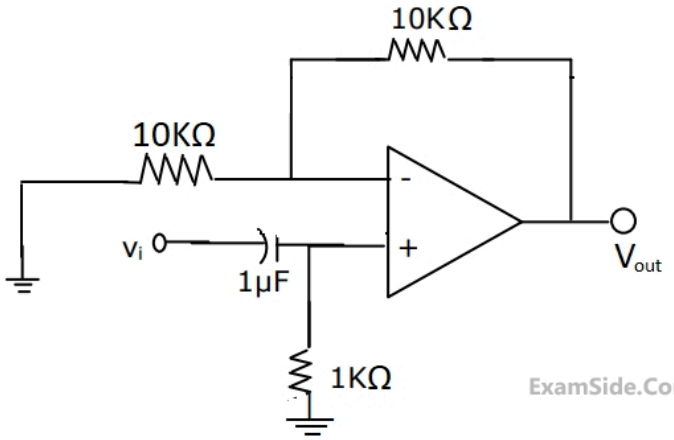
(D) Both the bias currents and the input offset current.

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Q73. GATE ECE 2005

Open solution

The Op-Amp circuit shown in figure is a filter. The type of filter and its cut-off frequency are respectively.



(A) high pass, 1000 rad/sec.

(B) low pass, 1000 rad/sec.

(C) high pass, 10 rad/sec.

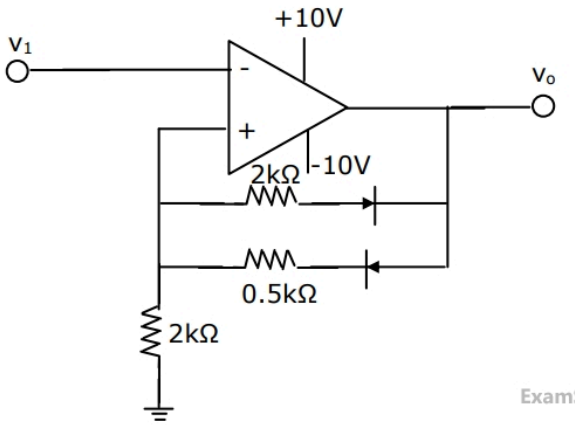
(D) low pass, 10 rad/sec.

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Q74. GATE ECE 2005

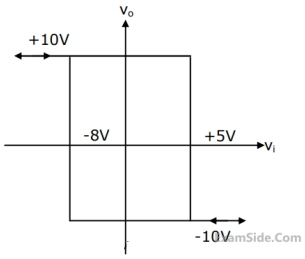
Open solution

Given the ideal operational amplifier circuit shown in figure, indicate the correct transfer characteristics assuming ideal diodes with zero cut-in voltage.

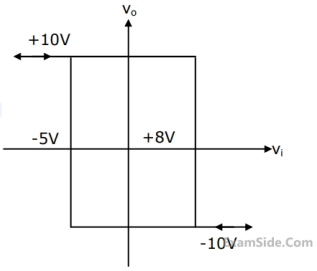


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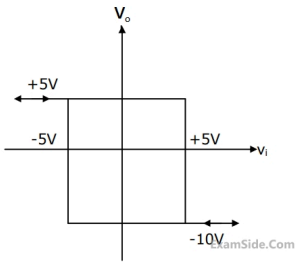
(A)



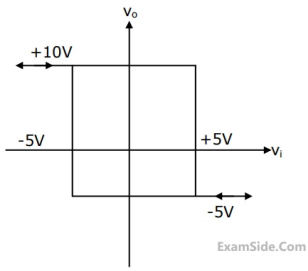
(B)



(C)



(D)



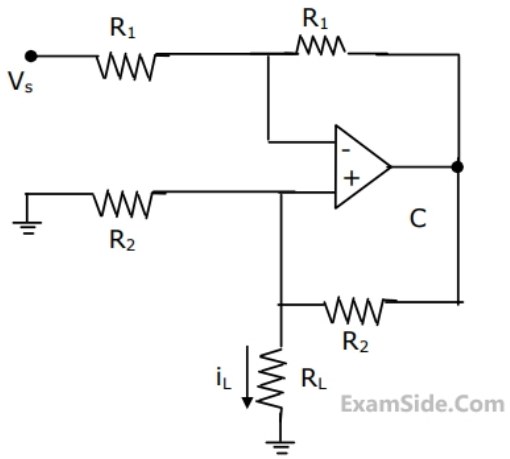
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Q75. GATE ECE 2004

Open solution

In the op-amp circuit given in the figure, the load current i_L is

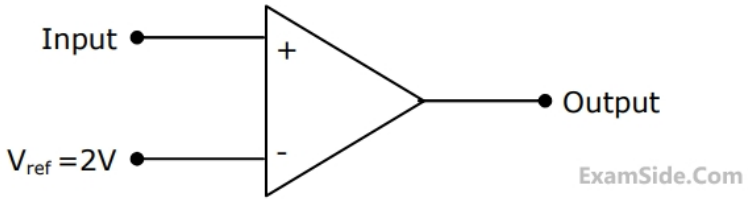


(A) $-\frac{V_s}{R_2}$ (B) $\frac{V_s}{R_2}$ (C) $\frac{V_s}{R_L}$ (D) $\frac{V_s}{R_1}$

Q76. GATE ECE 2003

Open solution

If the input to the ideal comparator shown in figure is sinusoidal signal of 8V (peak to peak) without any DC component, then the output of the comparator has a duty cycle of



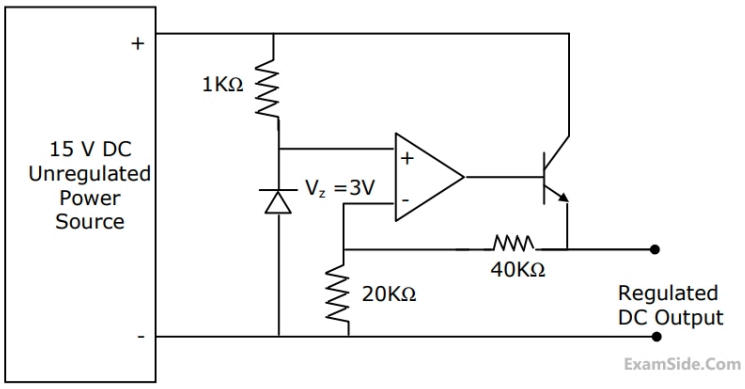
(A) 1/2 (B) 1/3 (C) 1/6 (D) 1/12

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Q77. GATE ECE 2003

Open solution

The output voltage of the Regulated power supply shown in the figure is

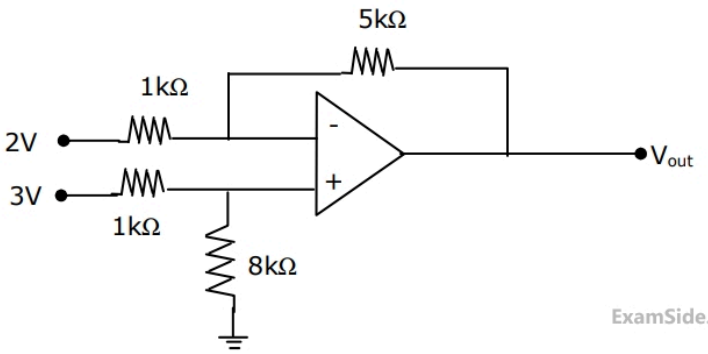


(A) 3 V (B) 6 V (C) 9 V (D) 12 V

Q78. GATE ECE 2003

Open solution

If the Op-Amp in the figure is ideal, the output voltage V_{out} will equal to



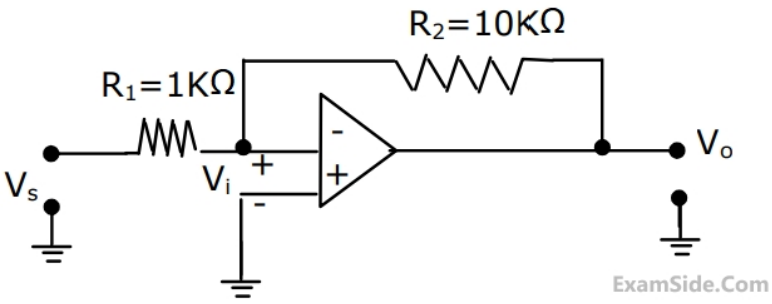
(A) 1 V (B) 6 V (C) 14 V (D) 17 V

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Q79. GATE ECE 2001

The Inverting Op-Amp shown in the figure, has an Open-loop gain of 100. The closed loop gain $\frac{V_o}{V_s}$ is

Open solution

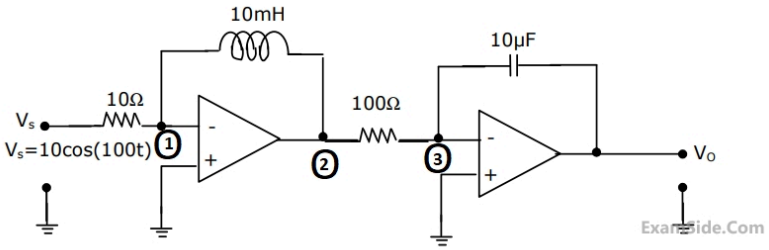


(A) -8 (B) -9 (C) -10 (D) -11

Q80. GATE ECE 2001

In the figure assume the Op-Amps to be ideal. The output V_0 of the circuit is

Open solution



(A) $10\cos(100\,t)$

(B) $10\int_0^t \cos\ (100\tau)\ d\tau$ V

(C) $10^{-4}\int_0^t \cos\ (100\tau)\ d\tau$

(D) $10^{-4}\frac{d}{dt}\cos(100t)$

Q81. GATE ECE 2001

An Amplifier using an Op-Amp with a Slew-Rate SR = 1 V/μsec, has a gain of 40 dB. If this amplifier has to faithfully amplify sinusoidal signals from dc to 20 kHz without introducing any Slew Rate induced distortion, then the input signal level must not exceed.

Open solution

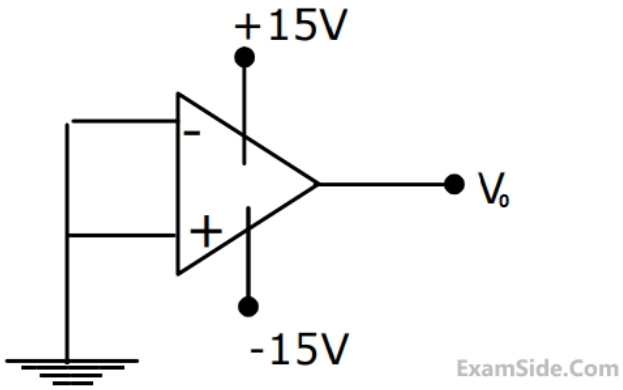
(A) 795 mV (B) 395 mV (C) 79.5 mV (D) 39.5 mV

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Q82. GATE ECE 2000

If the Op-Amp in the figure has an input offset voltage of 5 mV and an open-loop voltage gain of 10,000, V_0 will be

Open solution

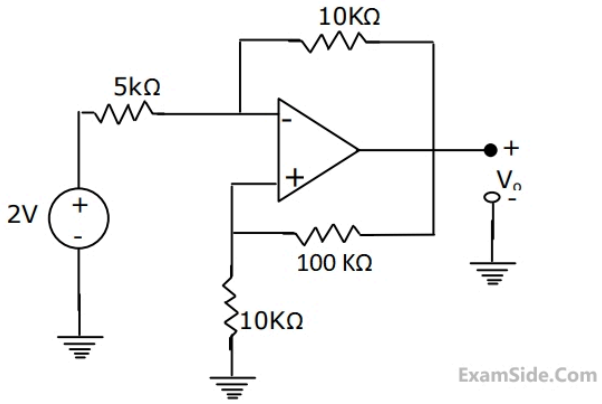


(A) 0 V
(B) 5 mV
(C) + 15 V (or) - 15V
(D) +50 V (or) -50V

Q83. GATE ECE 1997

The output voltage V_0 of the circuit shown in the figure is

Open solution



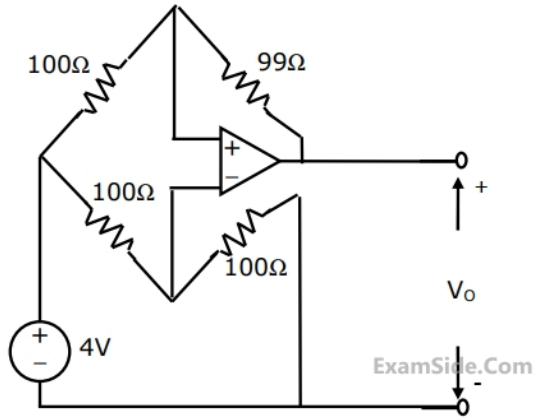
(A) -4 V
(B) 6 V
(C) 5 V
(D) -5.5 V

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Q84. GATE ECE 1993

For the ideal Op-Amp circuit of fig. Determine the output voltage V_0

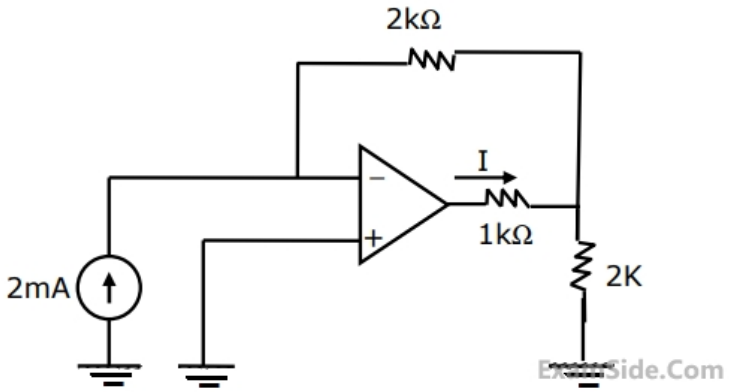
Open solution



Q85. GATE ECE 1992

Assume that the operational amplifier in figure is deal the current I through the **IK** ohm resistor is

Open solution

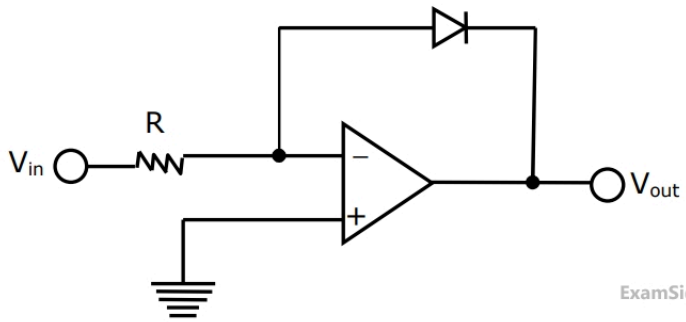


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Q86. GATE ECE 1992

The circuit of fig. uses an ideal OP_Amp for small positive values of V_{in}, the circuit works as

Open solution

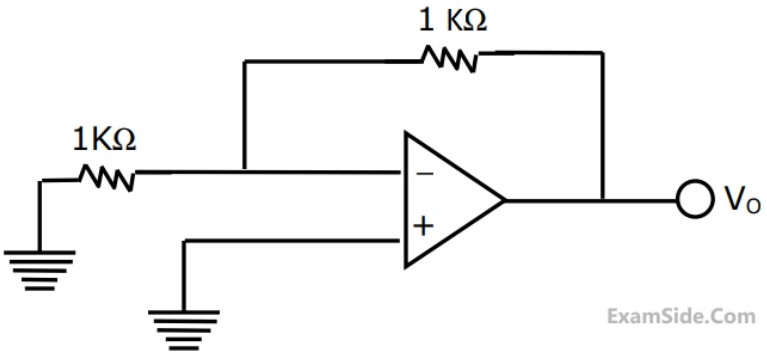


(A) a half wave rectifier
(B) a differentiator
(C) a logarithmic amplifier
(D) an exponential amplifier

Q87. GATE ECE 1992

An Op-Amp has an offset voltage of 1mV and is ideal in all other respects. If this OP-Amp is used in the circuit shown in fig. The O/P voltage will be (Select the nearest value).

Open solution



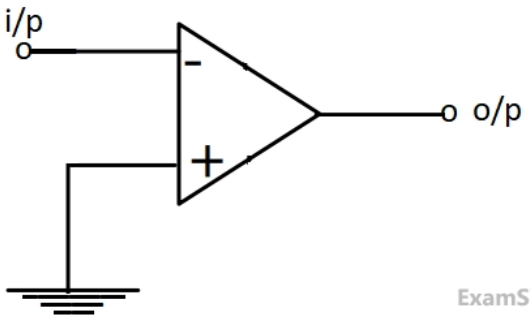
(A) 1 mV
(B) 1 V
(C) $\pm$ 2 mV
(D) 0 V

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Q88. GATE ECE 1990

If the input to the circuit of figure is a sine wave the output will be.

Open solution



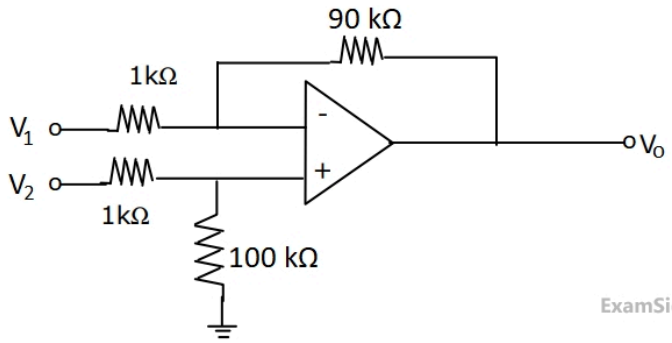
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(A) A half-wave rectified sine wave.
(B) A full-wave rectified sine wave.
(C) A triangular wave.
(D) A square wave.

Q89. GATE ECE 1990

The CMRR of the differential Amplifier of the fig is equal to

Open solution



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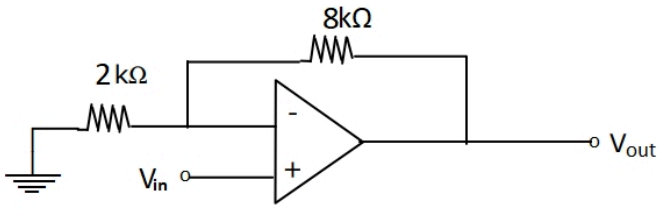
(A) 1000
(B) 0
(C) 900
(D) 1800

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Q90. GATE ECE 1990

Open solution

The Op-Amp of fig has a very poor open loop voltage gain of 45 but is otherwise ideal. The gain of the Amplifier equals:



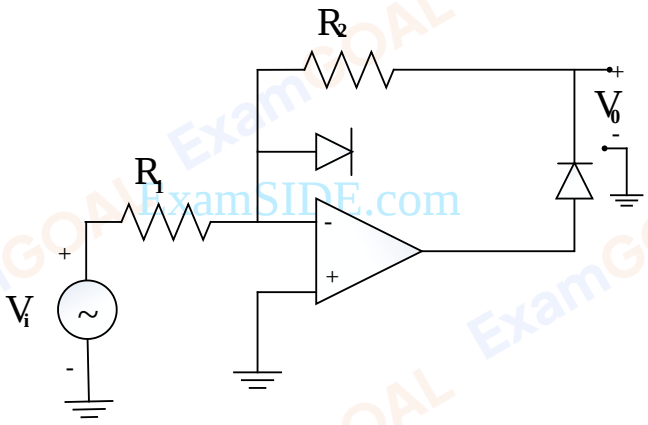
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(A) 5 (B) 20 (C) 4 (D) 4.5

Q91. GATE ECE 1989

Open solution

Refer to Fig.



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(A) For $V_i > 0,\ V_0 = -\frac{R_2}{R_1}V_i$

(B) For $V_i > 0,\ V_0 = 0.$

(C) For $V_i < 0,\ V_0 = -\frac{R_2}{R_1}V_i$

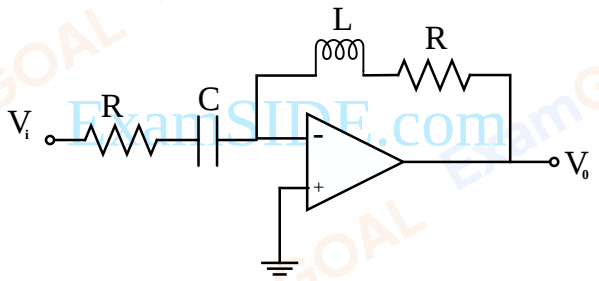
(D) For $V_i < 0,\ V_0 = 0.$

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Q92. GATE ECE 1988

Open solution

The OP-AMP shown in fig. below is ideal. $R = \sqrt{L/C}$. The phase angle between V_0 and V_i at $\omega = 1/\sqrt{LC}$ is



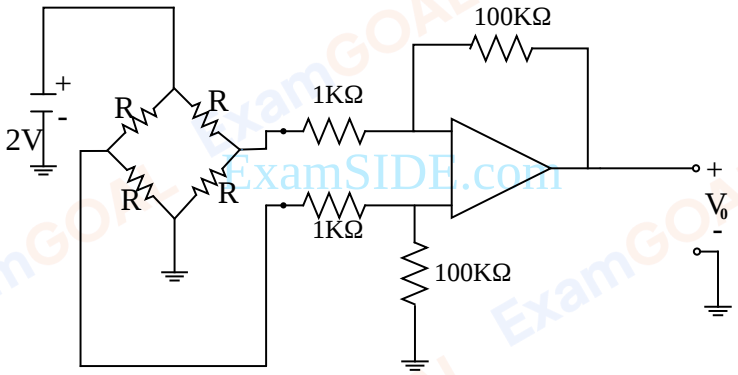
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(A) $\pi/2$ (B) π (C) $3\pi/2$ (D) 2π

Q93. GATE ECE 1987

Open solution

In Fig. if the CMRR of the operational amplifier is 60dB, then the magnitude of the output voltage is



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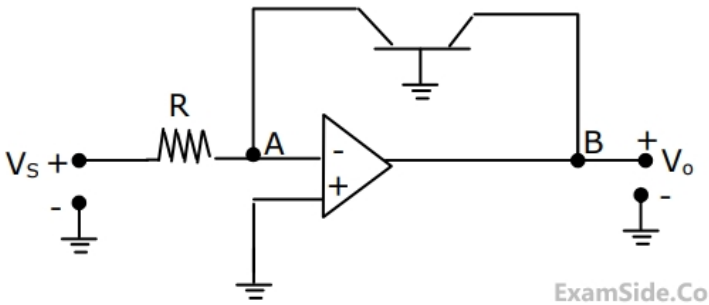
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Marks 5 6 questions

Q94. GATE ECE 2001

Open solution

Assume that the OP-AMP in the figure is ideal



(a) Obtain an expression for V_0 in terms of V_s, R, and the reverse saturation current I_s of the transistor.

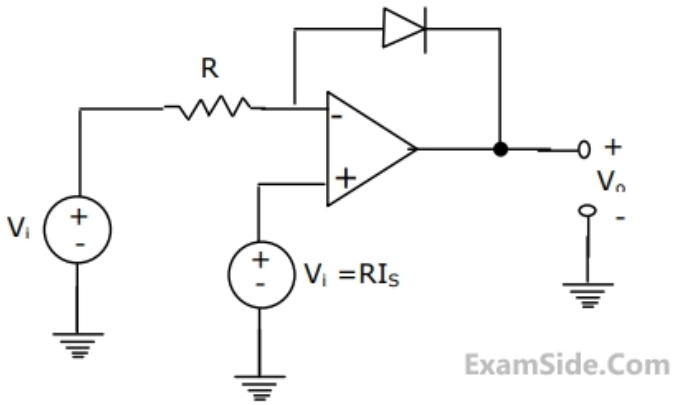
(b) If R = 1Ω, I_s = 1pA and the thermal voltage V_T = 25mV, then what is the value of the output voltage V_0 for an input voltage V_s = 1V?

(c) Suppose that the transistor in the feedback path is replaced by a p-n junction diode with a reverse saturation current of I_s. The p-side of the diode is connected to node A and the n-side to node B. Then what is the expression for V_0 in terms of V_s, R and I_s ?

Q95. GATE ECE 1997

Open solution

Consider the circuit given in fig. using an ideal operational Amplifier.



The characteristics of the diode are given by the relation $I = I_s \left[e^{\frac{qV}{KT}} - 1\right]$,

Where V is the forward voltage across the diode

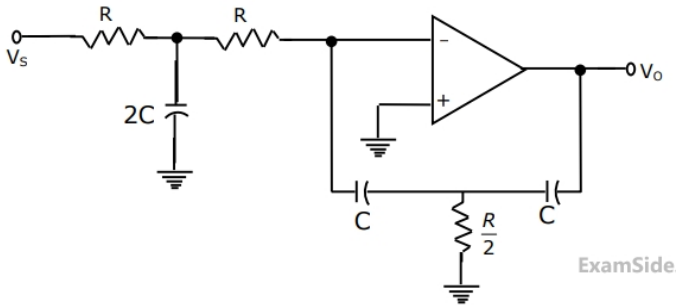
(a)Express V_0 as a function of V_i assuming $V_i > 0$.

(b)If R = 100KΩ , I_s = 1μA and $\frac{KT}{q} = 25mV$

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Q96. GATE ECE 1995

Shown that the system in Fig. is a double integrator. In other words, prove that the transfer gain is given by $\frac{V_0(s)}{V_i(s)} = \frac{1}{(CRs)^2}$, assume ideal OP -AMP

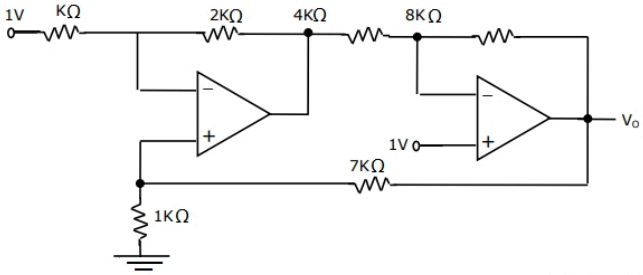


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Q97. GATE ECE 1994

Find the output voltage of the following circuit assuming ideal op-amp behavior.

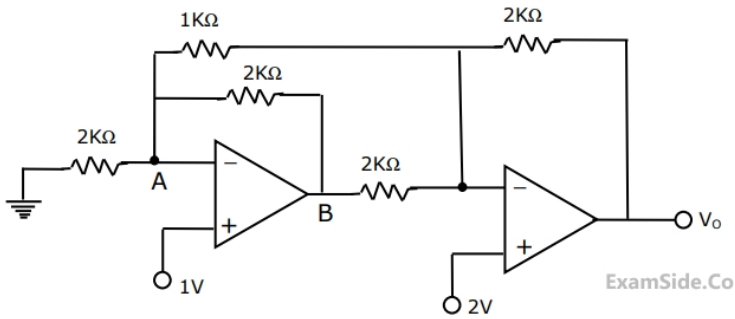


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Q98. GATE ECE 1993

Find the output voltage, V_0 in the following circuit assuming that the Op-Amps are ideal.



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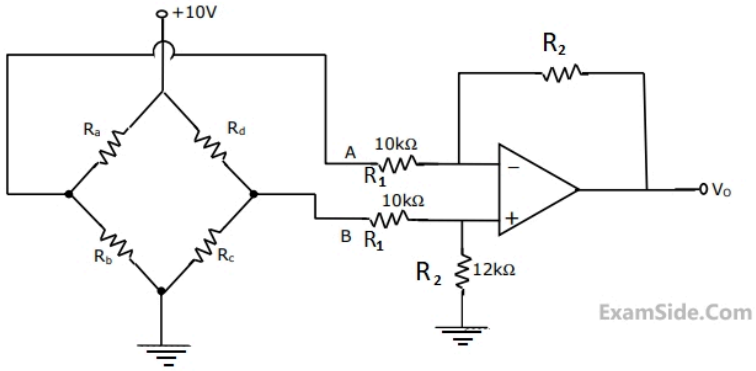
Q99. GATE ECE 1992

Open solution

Consider the circuit shown in Fig. The circuit uses an ideal operational amplifier. Assuming that the impedances at noodes A and B do not load the preceeding bridge circuit, calculate the output voltage V_o.

when $R_a = R_b = R_c R_d$ ohms.

when $R_a = R_b = R_c = 100$ ohms and $R_d = 120$ ohms.



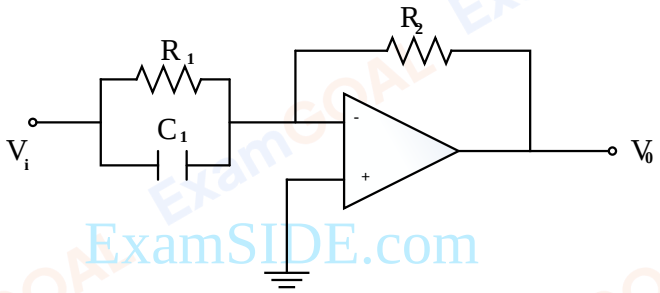
Marks 8 1 questions

Q100. GATE ECE 1988

Open solution

The Op-Amp in the Fig below is ideal. Find the output when V_i = (1-exp($-\alpha t$)) u(t). Assume the capacitor to be uncharged when the input is applied and $-\alpha$ = 1/CR_1.

$[u(t)\ is\ the\ unit\ step\ function.]$



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Answer Key

Marks 1

1. **C**	2. **D**	3. **A, C**	4. **B, C**	5. **D**
6. **C**	7. **-12 to -12**	8. **A**	9. **6**	10. **B**
11. **B**	12. **3**	13. **159.14**	14. **1.5**	15. **1**
16. **3.18**	17. **D**	18. **B**	19. **B**	20. **D**
21. **A**	22. **B**	23. **B**	24. **B**	25. **A**
26. **C**	27. **A**	28. **D**	29. **C**	30. **A**
31. **D**	32. **A**	33. **D**	34. **D**	35. **33 KΩ**
36. **250 KHz**	37. **B**	38. **Stability**		

Marks 2

39. **C**	40. **(2.90 to 3.10)**	41. **D**	42. **B**	43. **A**
44. **8.30 to 8.34**	45. **A**	46. **1.15**	47. **44 to 45**	48. **0.798**
49. **413.7**	50. **15**	51. **-800**	52. **-1**	53. **1**
54. **2.8**	55. **12**	56. **250**	57. **D**	58. **1093**
59. **C**	60. **D**	61. **C**	62. **B**	63. **B**
64. **B**	65. **C**	66. **C**	67. **D**	68. **A**
69. **C**	70. **D**	71. **D**	72. **C**	73. **A**
74. **B**	75. **A**	76. **B**	77. **C**	78. **B**
79. **B**	80. **A**	81. **C**	82. **C**	83. **D**
84. **0.02 V**	85. **- 4 mA**	86. **C**	87. **C**	88. **D**
89. **A**	90. **B**	91. **C**	92. **C**	93. **100 mV**

Marks 5

94. **(a)** $V_0 = V_T \ln \frac{I_s R}{V_s}$ **(b)** $V_0 = -0.69V$ **(c)** $V_0 = \pi V_T \ln \frac{I_s R}{V_s}$	95. **(a)** $V_0 = RI_s - V_T l_n \times 9\frac{V_i}{RI_s}$ **(b)** $V_i = 5.46\,V$	**Solve it.**	97. **V_0 = 4 V**	98. **V_0 = 6 V**
99. **(a)V_0 = 0V** **(b)V_0 = 0.545V**				

Marks 8

100. $V_0 = \frac{-R_2}{R_1} u(t)$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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